

Equation Sheet: Circuits

Intro Physics II • Lectures 14–17 • Knight ch. 26–28

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals.

Symbols and constants

I current (A = C/s)	$\mathcal{E}$ emf (V)	P power (W)
$V, \Delta V$ potential difference (V)	r internal resistance (Ω)	Q charge (C)
R resistance (Ω)	C capacitance (F)	τ time constant (s)
ρ resistivity ($\Omega\cdot\text{m}$)	L, A wire length / cross-section area	e 1.60×10^{-19} C

1. Current, resistance & Ohm's law (Knight 26.1–26.5, 27.1–27.4)

fundamental

Current — rate of charge flow $I = \frac{dQ}{dt} \xrightarrow{\text{steady}} I = \frac{Q}{\Delta t}$

Ohm's law
(for an ohmic resistor; also $\Delta V = IR$ across any R)

$$I = \frac{\Delta V}{R}$$

Resistor power
(rate electrical energy $\rightarrow$ heat/light)

$$P = I \Delta V = I^2 R = \frac{(\Delta V)^2}{R}$$

derived

Resistance of a uniform wire

$$R = \frac{\rho L}{A}$$

Conservation of charge at a wire

I is the same all along a series path

2. EMF & real batteries (Knight 28.2–28.3)

fundamental

EMF — energy supplied per unit charge $\mathcal{E} = \frac{dW}{dq}$ (units of volts)

derived

Terminal voltage of a real battery
(internal resistance r ; equals $\mathcal{E}$ only when $I = 0$)

$$\Delta V_{\text{term}} = \mathcal{E} - Ir$$

Current from a single-loop source

$$I = \frac{\mathcal{E}}{R + r}$$

3. Series & parallel combinations (Knight 28.4)

fundamental

Resistors in series

(same current; voltages add; R_{eq} is bigger than any one)

$$R_{\text{eq}} = R_1 + R_2 + \cdots$$

Resistors in parallel

(same voltage; currents add; R_{eq} is smaller than any one)

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$$

derived

Two resistors in parallel

$$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

Capacitors

(combine the opposite way to resistors)

$$C_{\text{par}} = C_1 + C_2 + \cdots, \quad \frac{1}{C_{\text{ser}}} = \sum \frac{1}{C_i}$$

Voltage divider (series R_1, R_2)

$$\Delta V_1 = \Delta V_{\text{tot}} \frac{R_1}{R_1 + R_2}$$

4. Kirchhoff's rules (Knight 28.1, 28.5)

fundamental

Junction rule (charge conservation)

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

Loop rule (energy conservation)

(+ across a source from - to +; $-IR$ across a resistor in the direction of I)

$$\sum_{\text{loop}} \Delta V = 0$$

derived

Ideal meters

ammeter: $R = 0$ (in series); voltmeter: $R = \infty$ (in parallel)

5. RC circuits (Knight 28.6)

fundamental

Capacitor relation

$$Q = C \Delta V_C$$

Time constant

$$\tau = RC$$

derived

Discharging (starts at Q_0, I_0)

(both fall to 37% after one τ)

$$Q(t) = Q_0 e^{-t/\tau}, \quad I(t) = I_0 e^{-t/\tau}$$

Charging through a battery $\mathcal{E}$

(reaches 63% of full charge after one τ ; I decays)

$$Q(t) = C\mathcal{E} \left(1 - e^{-t/\tau}\right)$$

Current is largest at $t = 0$

$$I_0 = \frac{\mathcal{E}}{R} \quad (\text{charging}); \quad I_0 = \frac{\Delta V_0}{R} \quad (\text{discharging})$$
